7 (a) (i) the wire cuts magnetic field B1

e.m.f. induced when there is a change/cutting of flux.................. B1

(ii) (Lenz) e.m.f. 'opposes' change causing it B1

as direction of movement changes, so does e.m.f. .... B1 [4]

(b) $x_0 = 1.5$ mV ...(allow ±0.1).. C1

$\omega = 2\pi / T = 2\pi / (3 \times 10^{-3})$ C1

$= 2090$ rad s^{-1} C1

$x = 1.5 \sin 2090t$ .... A1 [4]